

1. What is Python, and why is it popular?

- Python is a high level, general purpose programming and object oriented programming language. It is created by Guido van Rossum and released in 1991.
- It is used to web development server side and software development and mathematics and system scripting.
- It is very popular why it is Readability, versatility, ease of learning, lot of libraries, large community support.

2. What is an interpreter in Python?

- In Python, an interpreter is a program that reads and executes Python code line by line.
- It then translates the Python code into an intermediate representation called bytecode.
- Examples of interpreter Python or Python 3, VS code, Jupyter notebook, Google Colab etc..

3. What are pre-defined keywords in Python?

- Keywords are reserved words in Python that have special meaning and cannot be used as identifiers (variable names, function names, etc.).
- Examples of keywords include if, else, while, for, def, class, import, try, except, return, True, False, None, etc.
- Keywords are case-sensitive and cannot be redefined or overridden within a Python program.

4. Can keywords be used as variable names?

No, keywords cannot be used as variable names in Python.

5. What is mutability in Python?

- Mutable objects can be modified after creation.
- EX-list and dictionaries

Mutability affects how objects are stored and manipulated in memory.

6. Why are lists mutable, but tuples are immutable?

- lists are mutable because objects can be changed in after creation
- tuples are immutable because objects are cannot be changed in after creation.

7. What is the difference between “==” and “is” operators in Python?

- == operator checks for equality of value between two objects
- is operator checks for identity determining if two variables reference the exact same object in memory.

8. What are logical operators in Python?

- Logical operators in python are used to combine or modify boolean expressions True or False
- There are three logical operators in python
- and->Returns true if both operands are true otherwise false.
- or->Returns true if atleast one of the operands is true, returns false only if both are false.
- not-> Returns the opposite boolean value of the operand if the operands is true, it returns false.

9. What is type casting in Python?

- Type casting is referred as type conversion, is the process of changing the one data type in to another data type in python.

10. What is the difference between implicit and explicit type casting?

- Implicit type conversion is performed by a python interpreter only.
- explicit type conversion performed by the user .

11. What is the purpose of conditional statements in Python?

- The main purpose of the conditional statements in python control the flow of execution in a program.

- it will allow different blocks of code to be executed based on specific conditions met.

12. How does the elif statement work?

- The elif keyword in python stands for else if. it can be used in conditional statements to check for multiple conditions .

- for example if the first condition is false, it moves on to the next elif statement to check if that condition is true.

13. What is the difference between for and while loop?

- “for” loops are used for iterating over a sequence of elements.

- “while” loops are used for executing code until a specified condition becomes False.

14. Describe a scenario where a while loop is more suitable than a for loop

- A Scenario where a while loop is more suitable than a for loop when the number of iterations needed

is not known beforehand.